Alessio Salvatore

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Work Experience

Laelaps AI Jun 2026–Present
Robotics Engineer

- Orchestrating robot fleets: developing low-level skills and orchestrating them with AI agents; built from scratch the multi-agent RL patrol skill, from simulator to cloud training
- Defined metrics to evaluate the quality of the patrol, its adaptability to changes in the environment, and its unpredictability
- Deployed and evaluated the trained policies on Unitree Go2 quadrupeds, scoring my framework against state-of-the-art approaches; submitted as a paper to ICRA 2027

ETH Robotics Club Nov 2025–Aug 2026
Partnerships Team Lead. Closed the club's 2025–2026 partnerships, including the annual sponsor and Hack2026

JOIINT Lab — Italian Institute of Technology, Bergamo Repo Feb–Jun 2025
Research Intern — Industrial Robotics. Linear actuator CAD, real-time Arduino control, MATLAB validation

Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics
Coursework: Probabilistic AI, Advanced Machine Learning, Deep Generative Models, Robot Learning, Perception and Learning for Robotics, 3D Vision, Image Analysis and Computer Vision, Planning and Decision Making

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering
Final Grade: 110/110 *cum laude* (equivalent GPA ≈ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Publications

PRoMiSe: Persistent Multi-Robot Monitoring Sim-to-Real-to-Field Project Page Nov 2026
A. Salvatore, J. de los Rios, D. Rytz, M. Stamatopoulou — *CoRL 2026 Workshop Sim2Real2Field*, Austin, TX

Projects

Research Projects

Knowledge Distillation for Aerial Robots Control Video Report Feb–Jun 2026
Autonomous Systems Lab

- Model-based controllers for an overactuated drone (OMAV) lose accuracy when the vehicle drifts from its model, and disturbance observers can be too slow to compensate
- Distilled multiple expert controllers, each tuned to a different version of the vehicle, into a single recurrent neural controller that adapts from its recent flight history: in simulation it tracks velocity better and diverges far less under $\pm$30% model errors (41.5% $\rightarrow$ 24%), and it flew the OMAV without retraining

Goal-Driven Adaptive-Resolution 3D Mapping with RL Report Feb–Jun 2026
Computer Vision and Geometry Group

- A robot mapping a building at uniform voxel resolution spends most of its memory on regions irrelevant to its task
- Trained a PPO agent reallocating resolution per region online under a memory budget, on a sparse hierarchical grid in Habitat/HM3D: it keeps 79% of a uniformly fine map's point-goal success with 87% less memory on 20 unseen episodes

Selected Projects

Autonomous Exploration on a Unitree A2 Quadruped Repo 06/2026
ETH Robotics Summer School (ETHZ-RobotX) — deployed an autonomous search-and-explore mission with our team on a Unitree A2 at three field sites

Vision-Language-Action Policies on an SO-101 Arm Repo 05/2026
Project of the Class Robot Learning — fine-tuned SmolVLA for language-conditioned pick-and-place on an SO-101 arm at 10 FPS